ChatGPT - Choyan

Alright — I'll give you a **complete A→Z setup guide** for using **OpenGL with GLAD, GLFW, and GLM in VS Code (Windows)**.

Since you're using Windows + VSCode, this guide is for that.

✅ OVERVIEW OF WHAT WE'LL INSTALL

You need:

1. **MinGW-w64 (C++ compiler)**
2. **VS Code**
3. **C/C++ Extension**
4. **GLFW**
5. **GLAD**
6. **GLM**
7. Proper folder structure
8. Compile + Run commands

🟢 STEP 1 — Install MinGW-w64 (Compiler)

1️⃣ Download MinGW-w64

👉 Go here:
https://www.mingw-w64.org/downloads/

Click:

```
Code
Mingw-w64-builds
```

Direct link:
https://github.com/niXman/mingw-builds-binaries/releases

Download:

```
Code
```

```
x86_64-posix-seh
```

Latest version `.zip`

2️⃣ Extract It

Extract to:

```
C:\mingw64
```

Inside it you should see:

```
bin
include
lib
```

3️⃣ Add to PATH

Press:

```
Windows + R → sysdm.cpl
```

Advanced → Environment Variables

Under **System Variables → Path → Edit**

Add:

```
C:\mingw64\bin
```

Click OK.

4️⃣ Verify Installation

Open CMD:

Code

```
g++ --version
```

If it shows version → OK ✅

🟢 STEP 2 — Install VS Code

Download:

https://code.visualstudio.com/

Install normally.

🟢 STEP 3 — Install C++ Extension

Open VSCode → Extensions → Install:

Code

```
C/C++ by Microsoft
```

🟢 STEP 4 — Download GLFW

Go to:

https://www.glfw.org/download.html

Download:

Code

```
Windows pre-compiled binaries (64-bit)
```

Extract.

You'll see:

Code

```
include
lib-mingw-w64
```

🟢 STEP 5 — Download GLM

Go to:

https://github.com/g-truc/glm

Click:

```
Code → Download ZIP
```

Extract.

You only need the `glm` folder inside it.

🟢 STEP 6 — Generate GLAD

Go here:

https://glad.dav1d.de/

Settings:

```
Language: C/C++
Specification: OpenGL
API: OpenGL 4.6 (or 3.3)
Profile: Core
Generate Loader: ✔️
```

Click **Generate**

Download zip.

Extract.

You will get:

```
include/glad
src/glad.c
```

🟢 STEP 7 — Create Project Folder

Create folder:

```
OpenGL_Project
```

Inside it create:

```
OpenGL_Project/
 │
 ├─── include/
 │    ├─── glad/
 │    ├─── GLFW/
 │    └─── glm/
 │
 ├─── lib/
 │
 ├─── src/
 │    ├─── main.cpp
 │    └─── glad.c
```

Copy files like this:

From GLAD:

```
glad/include/glad → OpenGL_Project/include/glad
glad.c → OpenGL_Project/src/
```

From GLFW:

Copy:

```
GLFW/include/GLFW → OpenGL_Project/include/GLFW
GLFW/lib-mingw-w64/*.a → OpenGL_Project/lib/
```

From GLM:

Copy:

```
Code
glm folder → OpenGL_Project/include/
```

🟢 STEP 8 — Write Basic main.cpp

Put this inside `src/main.cpp`:

```C++
#include <glad/glad.h>
#include <GLFW/glfw3.h>
#include <iostream>

int main()
{
    if (!glfwInit())
        return -1;

    GLFWwindow* window = glfwCreateWindow(800, 600, "OpenGL", NULL, NULL);
    if (!window)
    {
        glfwTerminate();
        return -1;
    }

    glfwMakeContextCurrent(window);

    if (!gladLoadGLLoader((GLADloadproc)glfwGetProcAddress))
    {
        std::cout << "Failed to initialize GLAD\n";
        return -1;
    }

    while (!glfwWindowShouldClose(window))
    {
        glClear(GL_COLOR_BUFFER_BIT);
        glfwSwapBuffers(window);
        glfwPollEvents();
    }

    glfwTerminate();
```

```
    return 0;
}
```

🟢 STEP 9 — Compile Manually (Test First)

Open terminal inside project folder.

Run:

```
Code
g++ src/main.cpp src/glad.c -Iinclude -Llib -lglfw3 -lopengl32 -lgdi32 -o app.exe
```

If no errors → run:

```
Code
app.exe
```

You should see a window.

🟢 STEP 10 — Fix GLFW DLL Issue (Important)

If it says missing DLL:

Go inside GLFW download folder:

```
Code
lib-mingw-w64
```

Copy:

```
Code
glfw3.dll
```

Paste into:

```
Code
OpenGL_Project/
```

🟢 STEP 11 — Setup VS Code Auto Build (Optional but Recommended)

Create folder:

```
Code
.vscode
```

Inside create:

tasks.json

```JSON
{
  "version": "2.0.0",
  "tasks": [
    {
      "label": "build",
      "type": "shell",
      "command": "g++",
      "args": [
        "src/main.cpp",
        "src/glad.c",
        "-Iinclude",
        "-Llib",
        "-lglfw3",
        "-lopengl32",
        "-lgdi32",
        "-o",
        "app.exe"
      ],
      "group": {
        "kind": "build",
        "isDefault": true
      }
    }
  ]
}
```

Now press:

```
Code
Ctrl + Shift + B
```

It builds automatically.

🟢 STEP 12 — Enable GLM (Test)

Add in main.cpp:

```cpp
#include <glm/glm.hpp>
```

If no errors → GLM working ✅

🟢 STEP 13 — Common Problems Fix

If GLAD error:

Make sure:

```
src/glad.c
```

is included in compile command.

If undefined reference to glfw:

Make sure:

```
-Llib
-lglfw3
```

is correct.

🟢 FINAL PROJECT STRUCTURE

```
OpenGL_Project/
 |
```

```
├── include/
│   ├── glad/
│   ├── GLFW/
│   └── glm/
│
├── lib/
│   └── libglfw3.a
│
├── src/
│   ├── main.cpp
│   └── glad.c
│
├── glfw3.dll
├── app.exe
└── .vscode/
```

🟢 That's It

You now have:

✔️ OpenGL
✔️ GLAD
✔️ GLFW
✔️ GLM
✔️ VSCode build system

If you want, I can next give you:

- Proper CMake setup (cleaner method)
- Linux setup
- WSL setup
- Or how to structure a 3D engine properly

Just tell me.